

GIS-10 Water Shed

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1 Introduction

In this report, we will build a watershed model for a region of Vestland, Norway, using ArcGIS Pro. The main data source will be the N50 Kartdata, which provides both elevation and land cover information. We'll apply different thresholds on flow accumulation in an attempt to delineate the watershed boundaries similar to those present in the N50 Kartdata.

Topics we will explore:

- Filling sinks in the domain elevation model
- Calculating flow direction and flow accumulation
- Mask filtering with true and false values from rasters (or constant values)
- Delineating streams and watersheds from flow layers
- Comparing results with N50 Kartdata

2 Data

Elevation models for Norway are readily available from the Norwegian Mapping Authority (Kartverket). [N50 DTM](#) can be downloaded from the online catalog for specified cells across the country. If a user is interested in large areas, spanning multiple cells, the downloaded data can easily be merged into a single raster using ArcGIS Pro.

The [N50 Kartdata](#) is available from the same catalog. These data will be used to compare the resulting stream layers against, evaluating the thresholds we applied for the flow accumulation filtering.

The N50 Kartdata also contains information on lakes, which we will incorporate in our watershed model, to give a more hydrologically realistic representation of the watershed.

3 Methods

In class, we saw that the typical process for hydrological modeling is as follows:

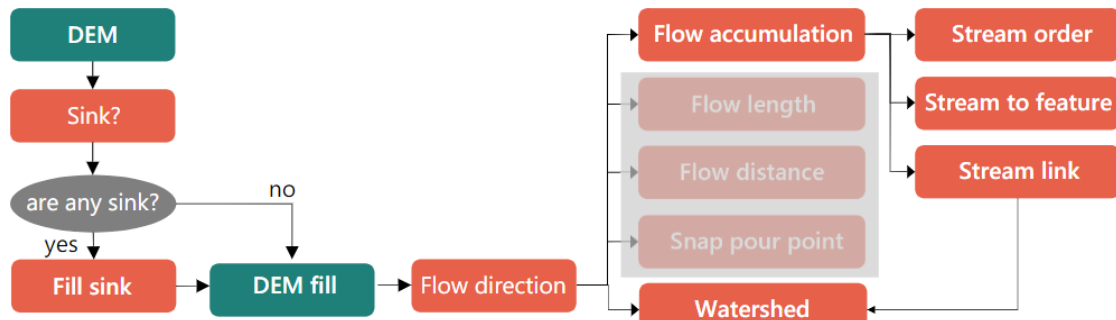


Figure 1:

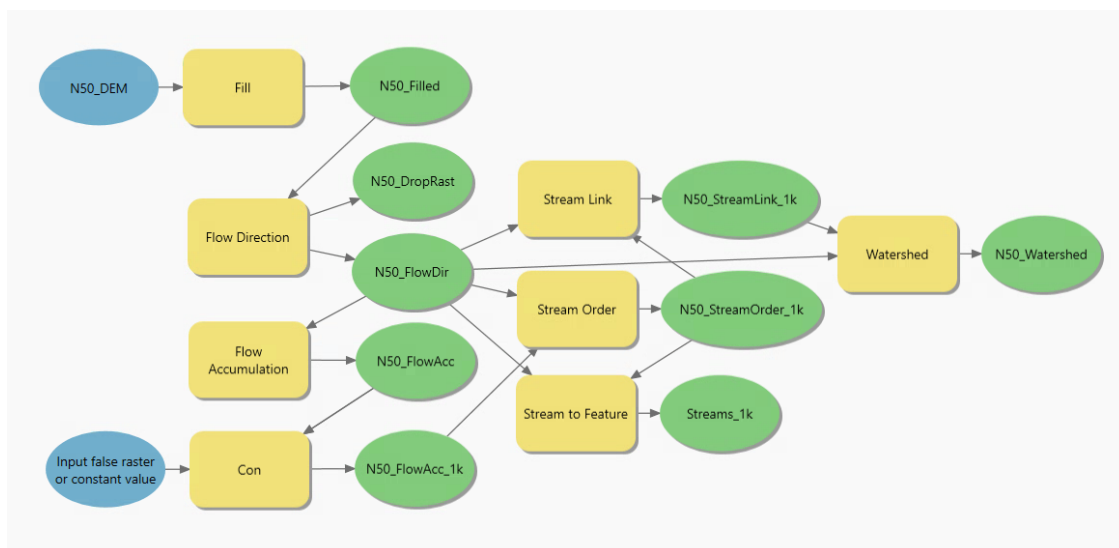
Typical hydrological modeling process. Source: [GEO4460 lecture notes](#).

For this lab exercise, we recreated most of these steps, focusing mainly on the flow accumulation and stream delineation steps.

Specifically, we:

- Filled sinks in the elevation model
- Calculated flow direction and flow accumulation
- Applied a mask to the flow accumulation layer to filter out low values
- (Optional) Subtracted the lakes from the flow accumulation layer to remove them from stream delineation
- Built a stream-order layer from the different filtered flow accumulation layers
 - The results here were compared against the N50 Kartdata stream layer to ensure our workflow was producing reasonable results
- Converted the streams to features and stream links, connecting smaller, lower order streams to larger, higher order streams or main rivers
- Calculated the watershed boundaries using the stream links and flow direction layers

The steps were configured through a model in ArcGIS Pro, allowing for easy adjustments and re-runs, which was particularly useful for testing different flow accumulation thresholds.



Figure

2: Our model workflow in ArcGIS Pro to build the watershed model.

3.1 Fill sinks and calculate flow

This report assumes we are using a precomputed and merged digital elevation model (DEM) of the area. We filled sinks in the DEM to ensure that all flow paths are directed towards lower elevations. This is crucial for accurate flow direction and accumulation calculations.

With the filled DEM, we calculated the flow direction using the **Flow Direction** tool in ArcGIS Pro. This tool computes the direction of flow for each cell in the raster, allowing us to understand how water would flow across the landscape. This means each cell is categorized with a direction: north, northeast, east, southeast, south, southwest, west, northwest, or no flow.

Next, we calculated the flow accumulation using the **Flow Accumulation** tool. This tool counts the number of upstream cells that contribute flow to each cell, resulting in a raster where each cell's value represents the total amount of flow that would pass through it. This tells us how much water would accumulate at each point in the landscape, which influences how large streams and rivers would become.

3.2 Filtering lakes and low flow accumulation values

Given that lakes are basically large accumulation areas, we can remove these from our flow accumulation and flow direction layers, labeling them as “no flow” areas. To remove lakes from the flow accumulation layer, we used the **Extract by Mask** tool in ArcGIS Pro. This tool allows us to extract the flow accumulation values that fall within the lake polygons, effectively masking out the lakes from the flow accumulation raster. We then set the flow accumulation values within the lake polygons to zero, indicating that no flow accumulates in these areas. (This step is optional. We can also choose to keep the lakes in the flow accumulation layer. More on each of these options in the results section.)

To filter away low flow accumulation areas, we used the **Con** tool in ArcGIS Pro, which allows us to apply a conditional statement to our flow accumulation raster. We set the condition to check if the cell value is greater than a specified threshold (e.g., 1000), and if true, we keep the original flow accumulation value; otherwise, we set it to zero.

We experimented with three thresholds: 500, 1000, and 10000. These numbers were selected based on the values from the flow accumulation raster, which ranged from 0 to 26000. The max values in that raster were clearly large rivers/fjord outlets, so the thresholds were set considerably lower to capture more of the smaller streams.

3.3 Stream delineation and stream order

With the filtered flow accumulation raster, we delineated streams using the **Stream Order** tool in ArcGIS Pro. This tool allows us to assign a stream order to each cell based on the flow accumulation values. Low order streams are small streams with low flow accumulation, while high order streams are larger rivers with high flow accumulation.

We used the **Stream Link** tool to convert the stream order raster into a vector layer, which connects smaller streams to larger ones. This step helped build a network of streams, where each stream segment is linked to its upstream and downstream neighbors.

3.4 Watershed Raster

Finally, we used the **Watershed** tool to delineate the watershed boundaries based on the stream links and flow direction layers. The inputs to this layer were the stream links and the flow direction raster. This tool identifies the area that drains into each stream segment, effectively delineating the watershed boundaries.

4 Results

4.1 Stream Orders

Figure 3: Stream orders for different flow accumulation thresholds (left to right: 0, 500, 1000, 10000). As the threshold increases, the number of streams remaining decreases, and the remaining streams become larger and more significant.

4.2 Water Shed Raster

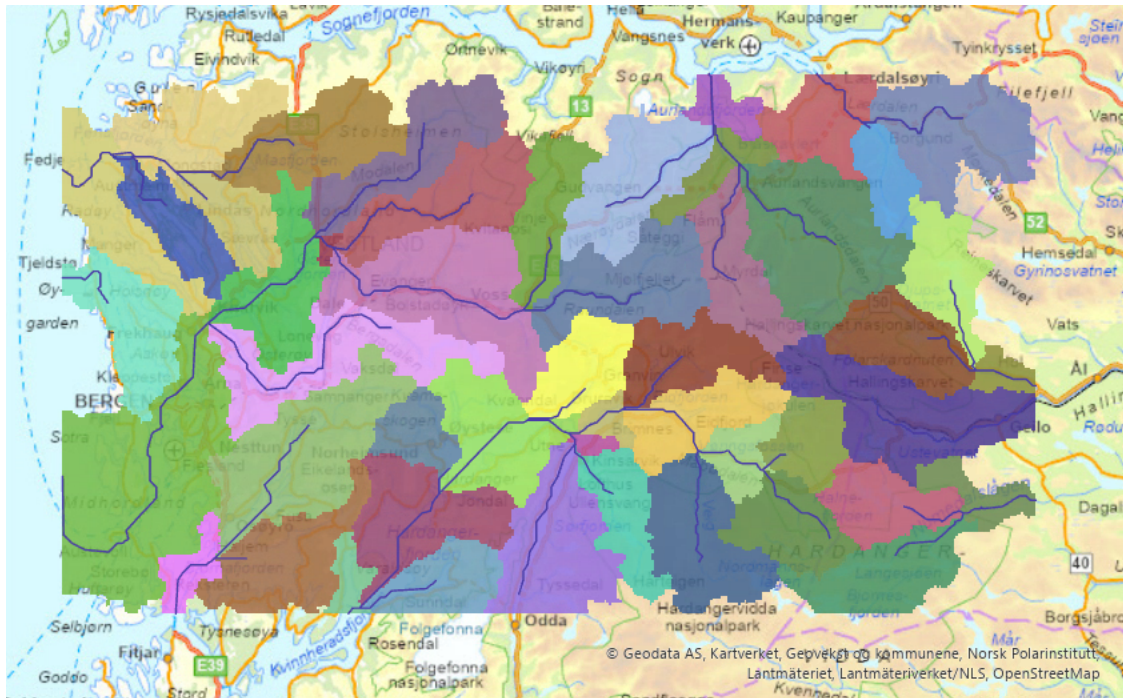


Figure 4:

Final watershed raster, showing the delineated watersheds based on the stream links and flow direction layers. Each color represents a different watershed area. Overlaid is the the streams layer generated from the 1000-thresholded stream links.

5 Discussion

5.1 Comparing Stream Orders

We compared stream orders with 3 different filter thresholds: 0 (unfiltered), 500, 1k, and 10k. The **unfiltered** stream order map shows a lot of streams scattered across the entire area observed.

The **500** and **1k** filters a moderate amount of streams. These seem to be the most meaningful results, as it shows stream along many of the main rivers, along with some smaller streams leading

into the main rivers. The **10k** filter shows only the biggest fjord outlets, which likely means the threshold is too high.

5.2 Comparing with N50 Kartdata

6 Conclusion/Summary

7 References

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